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Association between microinflammatory state and cardiac valvular calcification in patients undergoing maintenance hemodialysis: a single-centre observational study

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Abstract

Objective To investigate the association between microinflammatory state and cardiac valvular calcification (CVC) in maintenance hemodialysis (MHD) patients, and to compare the predictive value of the C-reactive protein-albumin-lymphocyte (CALLY) index with conventional inflammatory and nutritional biomarkers for CVC severity.

Methods This single-center cross-sectional study enrolled 203 MHD patients. Patients were categorized by echocardiography into no CVC ($n=67$), single valve calcification ($n=69$), and double valve calcification ($n=67$) groups. Serum levels of hs-CRP, IL-6, TNF-α, and routine biochemical parameters were measured. The CALLY index, neutrophil-to-lymphocyte ratio (NLR), C-reactive protein-to-albumin ratio (CAR), prognostic nutritional index (PNI), and geriatric nutritional risk index (GNRI) were calculated. Multinomial logistic regression, model fit indices, net reclassification improvement (NRI), integrated discrimination improvement (IDI), receiver operating characteristic (ROC) curves, and bootstrap internal validation were utilized to evaluate the comparative and incremental performance of the biomarkers.

Results The prevalence of CVC in the study population was 67.0%. Multinomial logistic regression analysis demonstrated that after adjusting for the six clinical covariates (age, gender, dialysis vintage, serum phosphorus, active vitamin D dosage, and phosphate binders), the CALLY index was independently associated with both single valve calcification (OR = 0.680, $P=0.005$) and double valve calcification (OR = 0.527, $P<0.001$). The CALLY index also exhibited the highest C-statistic for CVC (0.869). In NRI/IDI analyses, the CALLY index significantly improved both continuous NRI (0.435, $P=0.001$) and IDI (0.034, $P=0.004$) for double valve calcification and CVC, whereas NLR, IL-6, and hs-CRP did not.

Conclusion The CALLY index is independently associated with CVC severity in MHD patients. Incorporating the CALLY index into standard clinical parameters provides significant incremental predictive value for identifying multivalvular involvement.

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Keywords Cardiac valvular calcification, CALLY index, Inflammation, Maintenance hemodialysis

Introduction

Chronic kidney disease (CKD) has emerged as a major global health challenge, with an estimated 788 million adults aged 20 years and older affected worldwide in 2023, representing a global age-standardised prevalence of 14.2% [1]. The disease burden is particularly severe as it progresses to the end stage; according to the Global Burden of Disease Study 2023, the number of patients requiring kidney failure with replacement therapy, including maintenance hemodialysis, has risen to 4.59 million worldwide [2]. Cardiovascular disease remains the leading cause of mortality among patients with end stage kidney disease [3]. Cardiac valvular calcification, particularly involving the mitral and aortic valves, is a highly prevalent complication in these patients and serves as a strong marker for generalised atherosclerosis and adverse cardiovascular outcomes [4, 5]. CVC is an active and strictly regulated pathological process driven by metabolic derangements, oxidative stress, and chronic inflammation [6].

Microinflammation is defined as a persistent, low-grade systemic inflammatory state occurring independently of active infection, typically without overt clinical symptoms and evidenced only by mild elevations of circulating pro-inflammatory cytokines. First proposed by Schömig et al. in 2000, microinflammation is closely associated with protein-energy wasting in MHD patients [7]. Although its exact mechanisms remain incompletely understood, extensive research implicates metabolic disorders, oxidative stress, gut dysbiosis, and dialysate composition [8–10]. Mounting evidence suggests that microinflammation plays a pivotal role in the pathogenesis and progression of CVC [6, 11]. Specifically, inflammatory biomarkers such as high-sensitivity C-reactive protein (hs-CRP) and interleukin-6 (IL-6) have been strongly correlated with the development of vascular and valvular calcification [12–15]. The NLR, a simple marker reflecting systemic inflammatory state, has been demonstrated to be closely associated with cardiovascular calcification in MHD patients. Tang et al. reported that NLR was an independent risk factor for cardiovascular calcification and predicted CVC in a cohort of 362 MHD patients [16]. The CAR, which integrates inflammatory and nutritional state, has shown promising prognostic value in valvular heart disease. In a nationwide prospective cohort study of 5,519 patients with moderate-to-severe valvular heart disease, Zhang et al. demonstrated that a high CAR was independently associated with 2-year all-cause mortality and improved risk stratification when added to conventional risk models [17].

In the hemodialysis population, the interplay between inflammation, nutritional state, and cardiovascular outcomes has been increasingly recognized. Yaprak et al. demonstrated that the PNI exhibited superior predictive value for mortality compared with the NLR in a cohort of hemodialysis patients, underscoring the importance of composite inflammatory indices in this population [18]. Similarly, Wang et al., in a longitudinal study of hemodialysis patients, demonstrated that dynamic changes in the PNI and GNRI were significantly associated with reduced risks of both all-cause and cardiovascular mortality. Among the indices evaluated, the baseline GNRI and its post-dialysis trajectory exhibited the strongest predictive performance [19]. Furthermore, recent studies have emphasized the role of cardiometabolic risk profiling in dialysis patients, wherein multidimensional markers integrating inflammation and nutrition have shown incremental prognostic value beyond traditional risk factors [20]. These findings further illustrate the prognostic value of multidimensional indices that integrate nutritional and inflammatory parameters in the dialysis population.

The CALLY index is a novel integrative biomarker that reflects both nutritional and inflammatory state. Originally developed by Iida et al. based on the PNI to predict prognosis after hepatocellular carcinoma resection [21] its clinical utility has since expanded to tumor prognosis [22–24], the incidence of osteoarthritis [25], cardiovascular outcomes [26–28], and metabolic disorders [29, 30]. However, its prognostic value in MHD patients remains incompletely characterized. While Huang et al. reported that the CALLY index independently predicted all-cause and cardiovascular mortality in MHD patients [31, 32], Cakmak et al. found that it did not demonstrate independent prognostic value in a smaller cohort [33]. These discrepant findings underscore the need for further evaluation of the CALLY index in specific clinical contexts—including its potential association with cardiovascular structural abnormalities such as CVC, which has yet to be elucidated.

Despite these advances, studies specifically addressing the predictive value of NLR and CAR for CVC severity—particularly in distinguishing single versus double valve calcification—remain limited, and direct comparisons with novel composite markers such as the CALLY index are lacking.

Therefore, this study aimed to investigate the association between microinflammation and CVC in MHD patients, evaluate the predictive value of the CALLY index for CVC, and compare its performance with conventional inflammatory and nutritional biomarkers (hs-CRP, IL-6, NLR, PNI, GNRI, and CAR).

Materials and methods

Study population

A total of 203 patients undergoing MHD at the Blood Purification Centre of Xinhua Hospital, affiliated with Shanghai Jiao Tong University School of Medicine, were enrolled between December 2024 and December 2025. Among them, 120 (59.1%) were male and 83 (40.9%) were female.

Inclusion Criteria: (1) Aged ≥ 18 years; (2) On regular hemodialysis for ≥ 3 months, with a frequency of three sessions per week (4–5 h per session); (3) Provision of written informed consent.

Exclusion Criteria: (1) A history of acute infection within the last 6 months; (2) Severe cardiac, pulmonary, or hepatic dysfunction; (3) Active malignancy or currently undergoing oncology treatment; (4) Active phase of rheumatic or autoimmune diseases.

Research methods

Collection of clinical data

Demographic data, primary renal disease aetiology, medical history, and personal history were collected, including age, sex, dialysis vintage, height, weight, smoking history, and comorbidities such as hypertension, diabetes mellitus, and coronary heart disease.

Medication histories of all enrolled patients were reviewed. All patients were on long-term treatment with phosphate binders and active vitamin D. The type of phosphate binder (calcium-containing: calcium acetate, calcium carbonate D3; or non-calcium-containing: sevelamer carbonate, lanthanum carbonate chewable tablets) and the weekly dosage of active vitamin D were documented. Active vitamin D preparations included calcitriol and alfacalcidol, with dosages uniformly converted to equivalent calcitriol dose (μg / week) using the conversion formula: 1 μg calcitriol = 0.5 μg alfacalcidol. The use of calcimimetics was also recorded.

Biochemical and laboratory measurements

Pre-dialysis blood samples were collected to measure biochemical parameters, including alkaline phosphatase (ALP), serum albumin (Alb), serum prealbumin (PAB), blood urea nitrogen (BUN), serum creatinine (SCr), uric acid (UA), total cholesterol (TC), triglycerides (TG), high-density lipoprotein cholesterol (HDL-c), low-density lipoprotein cholesterol (LDL-c), intact parathyroid hormone (iPTH), serum calcium, phosphorus, and magnesium. Complete blood count parameters, comprising white blood cell count (WBC), platelet count (PLT), neutrophil count (N), lymphocyte count (L), and hemoglobin (Hb) were analysed along with inflammatory markers such as hs-CRP, IL-6, CRP, and tumour necrosis factor (TNF-α) by the Department of Clinical Laboratory at our hospital.

Body weight before and after blood purification, the duration of the blood purification session, and post-dialysis BUN were collected.

The following indicators were calculated:

CALLY index = [Albumin (g/L) × Lymphocyte count (10^9/L)]/[CRP (mg/L) × 10]

BMI = Body weight (kg) / Height (m^2)

NLR = Neutrophil count (10^9/L) / Lymphocyte count (10^9/L)

CAR = hs-CRP (mg/L) / Albumin (g/L)

PNI = Albumin (g/L) + 5 × Lymphocyte count (×10^9/L)

GNRI = 1.489 × serum albumin (g/L) + 41.7 × (current weight / ideal weight). Ideal body weight was calculated using the Lorenz formulae: for men, height (cm) – 100 – (height – 150)/4; for women, height (cm) – 100 – (height – 150) / 2.5.

Corrected calcium (Cac) = Serum calcium (mmol/L) + 0.02 × (40 − Albumin (g/L))

Calcium-phosphorus product (Ca×P) = Corrected calcium (mmol/L) × Serum phosphorus (mmol/L) × 12.4

$Kt/V = -\ln(C_{post} / C_{pre} - 0.008 \times t) + (4\text{-}3.5 \times C_{post} / C_{pre}) \times \Delta BW / BW_{post}$, where C_{post} and C_{pre} represent pre- and post-dialysis blood urea nitrogen concentrations, t denotes the dialysis session duration (hours), ΔBW indicates the intradialytic weight loss (kg), and BW_{post} is the post-dialysis body weight (kg)

Echocardiographic assessment

Colour Doppler echocardiography was performed by two designated experienced sonographers to evaluate and record calcification of the aortic and mitral valves. CVC was defined as the presence of at least one area of intense echogenicity (high-echo nodules) on the valve leaflets or annulus [34].

Definition and grouping

Coronary artery disease (CAD) was diagnosed according to the 2023 American Heart Association/American College of Cardiology (AHA/ACC) guidelines [35].

Diabetes mellitus (DM) was diagnosed according to the 2024 American Diabetes Association (ADA) criteria [36].

In the absence of a universally accepted threshold for defining microinflammation in maintenance hemodialysis patients, we avoided arbitrary dichotomization. Instead, circulating inflammatory biomarkers—hs-CRP, IL-6, and the derived indices (NLR, CAR, and CALLY index)—were analyzed as continuous variables in all regression models. For descriptive purposes, hs-CRP levels > 3 mg/L were considered indicative of elevated cardiovascular risk based on AHA/CDC guidelines, but this cutoff was not used to define groups for primary analyses [37].

Residual renal function (RRF) was evaluated based on a 24-hour urine output ≥ 100 mL, consistent with previous studies [38].

Patients were categorized based on echocardiographic findings:

(1) No CVC group: no calcification of either the aortic or mitral valve;
(2) Single valve calcification group (Single VC): calcification of either the aortic valve or the mitral valve only;
(3) Double valve calcification group (Double VC): calcification of both the aortic and mitral valves.

Blood purification modality and adequacy

All enrolled patients underwent blood purification three times weekly, consisting of two hemodialysis sessions and one hemodiafiltration session, with one session of hemodialysis combined with hemoperfusion performed every two weeks. Blood purification adequacy was assessed by the urea clearance index (Kt/V).

Statistical analysis

Analyses were conducted in SPSS 27.0 and R 4.5.3. Continuous variables were first evaluated for normality using the Shapiro-Wilk test. Normally distributed data were expressed as mean ± standard deviation ($\bar{x}\pm s$) and analyzed using Student's t-test or one-way ANOVA. Non-normally distributed data are expressed as median (interquartile range, IQR), and compared among the three CVC groups using the non-parametric Kruskal-Wallis H test. Categorical variables were summarized as counts (percentages) and compared using the Chi-square test or Fisher's exact test. Multivariable multinomial logistic regression was used to identify independent predictors of single and double valve calcification (reference: no CVC). Model discrimination was assessed using the C-statistic, and fit was evaluated by Nagelkerke's R^2 and the Akaike information criterion (AIC). Beyond comparing the area under the receiver operating characteristic curve (AUC), the relative parameters of continuous net reclassification improvement (NRI) and integrated discrimination improvement (IDI) were calculated to quantify the absolute reclassification and discrimination gains when adding each biomarker to the baseline clinical model. A two-tailed $P < 0.05$ was considered statistically significant.

Results

Baseline characteristics

A total of 203 MHD patients were enrolled, including 67 (33.0%) in the no CVC group, 69 (34.0%) in the single valve calcification group, and 67 (33.0%) in the double valve calcification group. As shown in Table 1, significant differences were observed among the three groups in age, dialysis vintage, non-calcium-containing

Table 1 Clinical characteristics of subjects in single valve calcification group, double valves calcification group and non-valve calcification group

Variable	No CVC (*n* = 67)	Single VC (*n* = 69)	Double VC (*n* = 67)	*P*-value
Gender (male; N, %)	40 (59.7%)	43 (62.3%)	37 (55.2%)	0.697
Age (years)	63.00 (54.00–69.00)	66.00 (61.00–73.00)	68.00 (63.00–75.00)	< 0.001*
Dialysis vintage (months)	41.00 (21.00–71.00)	60.00 (30.00–119.00)	61.00 (37.00–106.00)	0.003*
Current smokers (N, %)	10 (14.9%)	15 (21.7%)	6 (9.0%)	0.116
BMI (kg/ m^2)	21.80 (19.70–24.00)	21.60 (19.45–24.30)	21.90 (19.60–24.30)	0.951
Comorbidities (N, %)				
Hypertension	62 (92.5%)	64 (92.8%)	64 (95.5%)	0.719
Diabetes mellitus	23 (34.3%)	25 (36.2%)	25 (37.3%)	0.936
Coronary artery disease	19 (28.4%)	23 (33.3%)	21 (31.3%)	0.820
Etiology of CKD (N, %)				0.501
Hypertensive nephropathy	20 (29.9%)	16 (23.2%)	17 (25.4%)	0.379
Diabetic Nephropathy	17 (25.4%)	13 (18.8%)	18 (26.9%)	0.594
Nephrolithiasis	19 (28.4%)	25 (36.2%)	14 (20.9%)	0.433
Polycystic kidney disease	3 (4.5%)	3 (4.3%)	7 (10.4%)	0.522
Unknown	5 (7.5%)	4 (5.8%)	5 (7.5%)	0.301
Others	3 (4.5%)	8 (11.6%)	6 (9.0%)	0.244
Medication history				
Phosphate binder (non-calcium-containing)	52 (77.6%)	41 (59.4%)	24 (35.8%)	< 0.001*
Calcimimetics	32 (47.8%)	29 (42.0%)	32 (47.8%)	0.740
Active vitamin D dosage (μg / week)	1.75 (1.75–3.50)	1.75 (0.88–1.75)	1.75 (0.88–1.75)	< 0.001*
Residual renal function	11 (16.4%)	7 (10.1%)	9 (13.4%)	0.559

Note: *$P < 0.05$; statistically significant. BMI, body mass index

phosphate binder use, and active vitamin D dosage (all $P<0.05$). Patients in the double valve calcification group were older, had longer dialysis vintage, lower rates of non-calcium-containing phosphate binder use, and lower active vitamin D dosage. No significant differences were found in sex, BMI, smoking history, or comorbidities (all $P>0.05$).

Laboratory parameters

The CALLY index in the overall cohort ranged from 0.12 to 8.03, with a median of 1.04 (IQR: 0.6–2.07). The distribution was right-skewed (skewness = 1.689). Significant differences were observed in albumin, serum phosphorus, calcium-phosphorus product, iPTH, hs-CRP, IL-6, TNF-α, NLR, CAR, and the CALLY index (all $P<0.05$). Notably, the CALLY index showed a stepwise decrease with increasing CVC severity ($P<0.001$), while inflammatory markers showed a stepwise increase. The detailed laboratory results and echocardiographic findings are summarized in Table 2. The distributions of the CALLY index, NLR and CAR were analyzed across different categories of CVC groups (Figs. 1, 2, and 3).

Independent risk factors for CVC

Univariate analysis revealed that age, dialysis vintage, albumin, serum phosphorus, calcium-phosphorus

Table 2 Laboratory and echocardiographic parameters of the study population

Variable	No CVC ($n=67$)	Single VC ($n=69$)	Double VC ($n=67$)	*P*-value
WBC (10^9/L)	5.84 (4.64–6.97)	5.80 (4.90–7.39)	5.79 (4.69–7.08)	0.628
Neutrophils (×10^9/L)	3.85 ± 1.22	4.21 ± 1.23	4.12 ± 1.30	0.222
Lymphocytes (×10^9/L)	1.19 ± 0.41	1.07 ± 0.38	1.03 ± 0.38	0.053
Hb (g/L)	112.40 ± 13.05	114.88 ± 13.92	115.16 ± 13.04	0.420
PLT (10^9/L)	172.67 ± 59.02	187.01 ± 52.28	168.76 ± 66.29	0.170
NT-proBNP (pg/ml)	3275.00 (1424.08–6015.42)	3497.74 (1529.93–7246.99)	4568.00 (2870.00–9868.40)	0.009*
Scr (μmol/L)	908.00 (754.00–1073.00)	919.00 (780.50–1121.00)	920.00 (742.00–1028.00)	0.628
BUN (mmol/L)	26.50 ± 6.39	25.87 ± 7.16	26.41 ± 6.97	0.846
Kt/V	1.41 (1.24–1.61)	1.43 (1.24–1.63)	1.46 (1.27–1.68)	0.791
UA (μmol/L)	457.00 (392.00–496.00)	404.00 (363.00–466.00)	442.00 (357.00–490.00)	0.057
PAB (mg/L)	315.67 ± 107.19	308.25 ± 84.21	295.79 ± 88.71	0.465
Alb (g/L)	39.20 (37.20–41.60)	37.30 (34.90–41.20)	37.60 (33.80–39.50)	< 0.001*
TG (mmol/L)	1.59 (0.93–2.68)	1.51 (1.01–2.32)	1.53 (1.12–2.25)	0.916
TC (mmol/L)	3.51 (3.00–4.38)	3.59 (3.03–4.27)	3.40 (2.81–4.01)	0.632
HDL-c (mmol/L)	0.92 (0.67–1.08)	0.91 (0.74–1.08)	0.89 (0.69–1.19)	0.992
LDL-c (mmol/L)	2.03 (1.66–2.52)	2.02 (1.56–2.81)	1.88 (1.43–2.42)	0.685
Calcium (mmol/L)	2.17 (2.05–2.33)	2.18 (2.04–2.39)	2.15 (2.05–2.39)	0.905
Cac (mmol/L)	2.16 (2.05–2.33)	2.22 (2.12-2.38)	2.25 (2.10-2.40)	0.204
Phosphorus (mmol/L)	1.69 ± 0.48	2.01 ± 0.61	2.03 ± 0.50	< 0.001*
Ca×P	45.93 ± 13.69	55.83 ± 17.71	56.79 ± 15.05	< 0.001*
Magnesium (mmol/L)	1.06 (0.97–1.19)	1.05 (0.94–1.17)	1.02 (0.95–1.15)	0.511
iPTH (pmol/L)	22.09 (13.12–38.80)	29.05 (14.81–48.52)	41.22 (18.55–80.56)	0.003*
ALP (U/L)	85.00 (63.00–123.00)	88.00 (64.00–112.00)	96.00 (69.00–120.00)	0.382
LAD (mm)	37.50 (33.70–39.30)	37.50 (34.40–42.45)	40.00 (37.40–44.20)	< 0.001*
LVEF (%)	65.00 (62.30–67.30)	65.90 (62.75–67.70)	65.20 (63.30–67.70)	0.978
hs-CRP (mg/L)	2.37 (0.90–4.55)	3.71 (2.27–5.91)	3.96 (3.04–5.87)	< 0.001*
TNF-α (pg/ml)	17.20 (12.40–22.70)	22.00 (18.15–28.66)	21.20 (17.50–29.20)	< 0.001*
IL–6 (pg/ml)	3.88 (2.89–6.95)	6.08 (4.57–8.43)	6.74 (4.61–10.50)	< 0.001*
NLR	3.11 (2.50–4.48)	3.89 (3.00–5.29)	4.17 (3.10–5.60)	0.004*
CAR	0.06 (0.03–0.13)	0.11 (0.07–0.21)	0.13 (0.09–0.20)	< 0.001*
CALLY index	1.84 (1.02–3.75)	0.97 (0.46–1.67)	0.67 (0.48–1.41)	< 0.001*
PNI	45.25 (42.50–48.65)	43.60 (39.33–46.60)	41.80 (39.10–48.06)	< 0.001*
GNRI	98.58 (95.13–103.20)	95.60 (90.23–100.48)	94.59 (91.28–102.18)	0.002*

Note: *$P<0.05$; statistically significant. ALP, alkaline phosphatase; Alb, albumin; PAB, serum prealbumin; BUN, blood urea nitrogen; SCr, serum creatinine; UA, uric acid; TC, total cholesterol; TG, triglycerides; HDL-c, high-density lipoprotein cholesterol; LDL-c, low-density lipoprotein cholesterol; WBC, white blood cell count; PLT, platelet count; Hb, hemoglobin; hs-CRP, high-sensitivity C-reactive protein; IL-6, interleukin-6; TNF-α, tumour necrosis factor-α; NT-proBNP, N-terminal pro-brain natriuretic peptide; iPTH, intact parathyroid hormone; LAD, left atrial diameter; LVEF, left ventricular ejection fraction; CALLY index, C-reactive protein-albumin-lymphocyte index; NLR, neutrophil-to-lymphocyte ratio; CAR, C-reactive protein-to-albumin ratio; Cac, Corrected calcium; Ca×P, Calcium-phosphorus product; PNI, prognostic nutritional index; GNRI: geriatric nutritional risk index

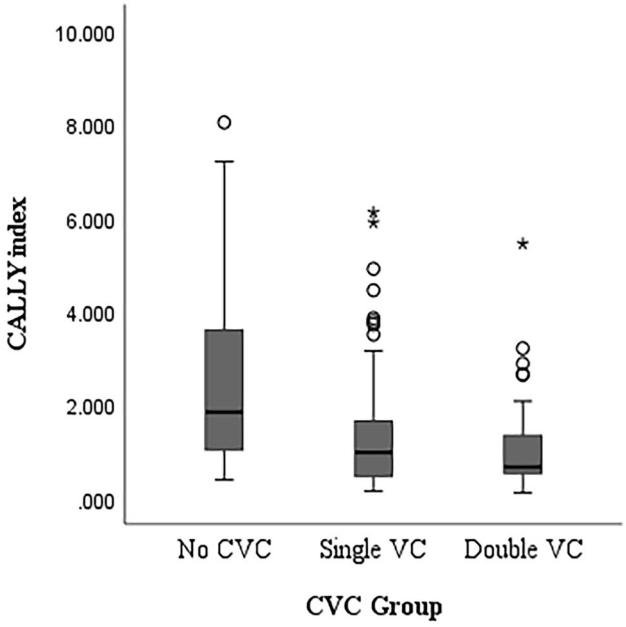


Fig. 1 Distribution of CALLY index by CVC Group. This figure presents a box-plot showing the distribution of the CALLY index across three groups of patients: no cardiac valve calcification (No CVC), single valve calcification (Single VC), and double valve calcification (Double VC)

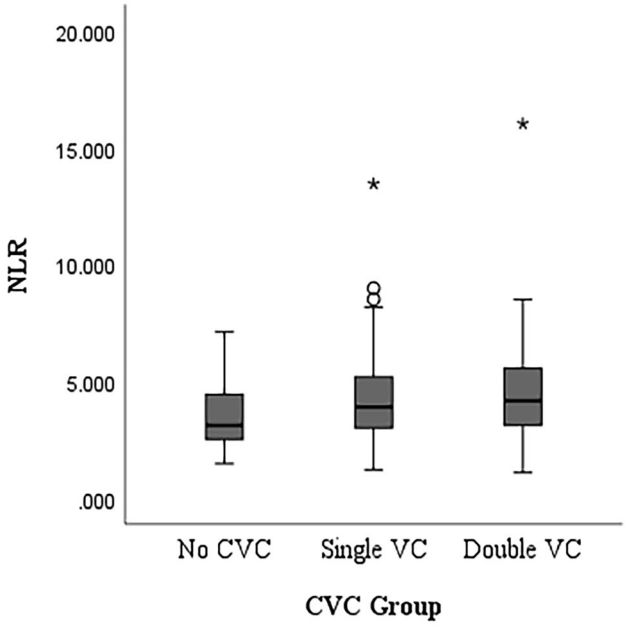


Fig. 2 Distribution of NLR by CVC Group. This figure displays a box-plot of the NLR according to the three CVC groups: No CVC, Single VC, and Double VC

product, iPTH, hs-CRP, IL-6, NLR, CAR, and LAD were significantly associated with CVC (all $P<0.05$). The CALLY index was significantly protective against both single valve (OR = 0.645, $P<0.001$) and double valve calcification (OR = 0.456, $P<0.001$) (Table 3).

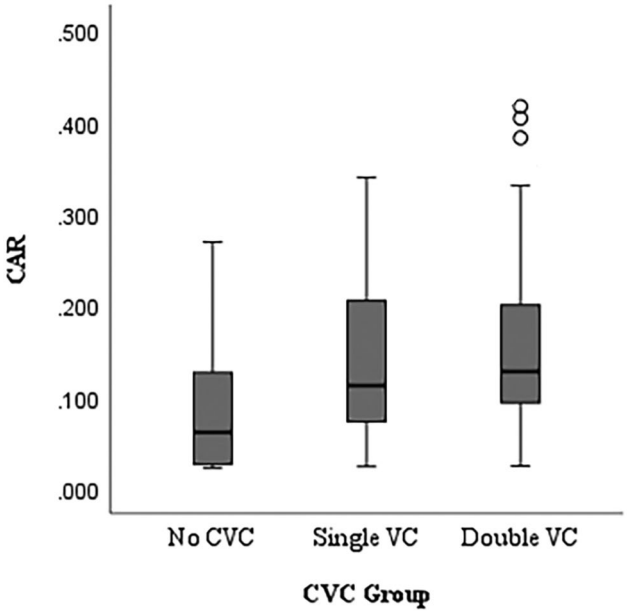


Fig. 3 Distribution of CAR by CVC Group. This figure illustrates a box-plot of the CAR across the three CVC groups: No CVC, Single VC, and Double VC

After adjusting for potential confounders, independent risk factors for double valve calcification included age (OR = 1.119, $P<0.001$), dialysis vintage (OR = 1.014, $P=0.001$), serum phosphorus (OR = 5.880, $P<0.001$), Ca×P (OR = 1.051, $P<0.001$), and LAD (OR = 1.101, $P=0.028$). Non-calcium-containing phosphate binder use (OR = 0.175, $P<0.001$) and active vitamin D dosage (OR = 0.536, $P=0.002$) were protective factors. (Table 4).

To evaluate the independent statistical associations between these continuous biomarkers and CVC severity, multinomial logistic regression models were executed using the "no CVC" group as the reference. To control for potential clinical confounding without introducing a risk of model overfitting, a fully adjusted 7-variable regression model was strictly applied for each biomarker. The baseline adjustment block comprised six pre-specified covariates: age, sex, dialysis vintage, serum phosphorus, active vitamin D dosage, and non-calcium-based phosphate binders.

As presented in Table 5, after controlling for the six baseline clinical covariates, the continuous CALLY index remained significantly and independently associated with both single valve calcification (OR = 0.680, 95% CI: 0.518–0.892, $P=0.005$) and double valve calcification (OR = 0.527, 95% CI: 0.370–0.749, $P<0.001$). This model also demonstrated the best goodness-of-fit, achieving the highest Nagelkerke R^2 (0.455) and the lowest Akaike Information Criterion (AIC) (372.913). In comparison, although CAR, NLR, IL-6, PNI and hs-CRP were significantly associated with CVC, their model fit indices were inferior (Table 5).

Table 3 Univariate logistic regression analysis of cardiac valve calcification

Variable	Single VC ($n=69$)		Double VC ($n=67$)	
	OR (95% CI)	*P*-value	OR (95% CI)	*P*-value
Age (years)	1.040 (1.006–1.075)	0.019*	1.070 (1.032–1.110)	< 0.001*
Dialysis vintage (months)	1.006 (0.999–1.012)	0.107	1.008 (1.002–1.015)	0.014*
NT-proBNP (pg/ml)	1.000 (1.000–1.000)	0.087	1.000 (1.000–1.000)	0.041*
Alb (g/L)	0.905 (0.824–0.993)	0.034*	0.842 (0.763–0.929)	< 0.001*
Phosphorus (mmol/L)	3.126 (1.595–6.127)	< 0.001*	3.389 (1.715–6.697)	< 0.001*
Ca×P	1.044 (1.019–1.069)	< 0.001*	1.048 (1.023–1.073)	< 0.001*
iPTH (pmol/L)	1.011 (1.000–1.022)	0.056	1.017 (1.006–1.028)	0.002*
hs-CRP (mg/L)	1.327 (1.135–1.550)	< 0.001*	1.365 (1.167–1.598)	< 0.001*
TNF-α (pg/ml)	1.003 (0.996–1.009)	0.398	1.002 (0.996–1.009)	0.544
IL−6 (pg/ml)	1.174 (1.061–1.299)	0.002*	1.188 (1.074–1.315)	< 0.001*
NLR	1.366 (1.100–1.696)	0.005*	1.403 (1.129–1.744)	0.002*
CAR	3.054 (1.791–5.207)	< 0.001*	3.509 (2.048–6.012)	< 0.001*
CALLY index	0.645 (0.510–0.817)	< 0.001*	0.456 (0.325–0.638)	< 0.001*
PNI	1.002 (0.981–1.023)	0.866	0.900 (0.841–0.964)	0.003*
GNRI	1.003 (0.988–1.017)	0.731	0.947 (0.906–0.990)	0.017*
LAD (mm)	1.035 (0.969–1.105)	0.305	1.130 (1.055–1.211)	< 0.001*
Phosphate binder (non-calcium-containing)	0.422 (0.200–0.893)	0.024*	0.161 (0.075–0.345)	< 0.001*
Active vitamin D dosage (μg/week)	0.574 (0.415–0.795)	< 0.001*	0.583 (0.421–0.808)	0.001*

Note: *$P < 0.05$; statistically significant. Alb, albumin. CAR was analyzed after being standardized for each increase of 0.1

Table 4 Multivariate logistic regression analysis of cardiac valve calcification factors

Variable	Single VC ($n=69$)		Double VC ($n=67$)	
	OR (95% CI)	*P*-value	OR (95% CI)	*P*-value
Age (years)	1.078 (1.032–1.125)	< 0.001*	1.119 (1.064–1.177)	< 0.001*
Dialysis vintage (months)	1.010 (1.002–1.018)	0.013*	1.014 (1.006–1.023)	0.001*
Phosphate binder (non-calcium-containing)	0.430 (0.187–0.993)	0.048*	0.175 (0.071–0.428)	< 0.001*
Phosphorus (mmol/L)	5.017 (2.193–11.475)	< 0.001*	5.880 (2.372–14.578)	< 0.001*
Ca×P	1.040 (1.015–1.066)	0.002*	1.051 (1.022–1.081)	< 0.001*
iPTH (pmol/L)	1.008 (0.992–1.023)	0.332	1.010 (0.994–1.026)	0.213
LAD (mm)	0.994 (0.918–1.076)	0.878	1.101 (1.011–1.200)	0.028*
Active vitamin D dosage (μg/week)	0.550 (0.381–0.794)	0.001*	0.536 (0.359–0.799)	0.002*

Note: *$P < 0.05$; statistically significant

Table 5 Model fit indices for different biomarkers in predicting CVC

Model	Variable	Single VC ($n=69$)		Double VC ($n=67$)		Nagelkerke R^2	AIC
		OR (95% CI)	*P*-value	OR (95% CI)	*P*-value		
Base Model	-	-	-	-	-	0.394	386.515
Model 1	CALLY index	0.680 (0.518–0.892)	0.005*	0.527 (0.370–0.749)	< 0.001*	0.455	372.913
Model 2	CAR	2.470 (1.330–4.587)	0.004*	2.529 (1.323–4.835)	0.005*	0.431	379.884
Model 3	NLR	1.384 (1.064–1.801)	0.015*	1.431 (1.089–1.881)	0.010*	0.423	382.376
Model 4	IL−6	1.123 (1.002–1.258)	0.046*	1.130 (1.008–1.267)	0.037*	0.413	385.212
Model 5	hs-CRP	1.208 (1.007–1.448)	0.041*	1.193 (0.983–1.447)	0.074	0.411	385.835
Model 6	PNI	0.999 (0.974–1.025)	0.959	0.916 (0.845–0.993)	0.032*	0.416	384.207
Model 7	GNRI	1.001 (0.982–1.021)	0.913	0.961 (0.911–1.013)	0.134	0.408	386.697

Note: *$P < 0.05$; statistically significant. All models adjusted for age, sex, dialysis vintage, serum phosphorus, phosphate binder type, and active vitamin D dosage. CAR was standardized per 0.1-unit increase

Table 6 presents the C-statistics for predicting the presence of any CVC. The CALLY index model achieved the highest C-statistic (0.869), followed by NLR (0.861), CAR (0.859), IL-6 (0.852), hs-CRP (0.850), and PNI (0.844). All models significantly discriminated between patients with and without CVC (all $P < 0.001$).

As calculated via the continuous metrics, the addition of the CALLY index achieved a statistically significant continuous net reclassification improvement

Table 6 C-statistics for predicting the presence of any CVC

Model	Variable	AUC (95% CI)	*P*-value
Base Model	-	0.844 (0.791–0.896)	< 0.001*
Model 1	CALLY index	0.869 (0.822–0.916)	< 0.001*
Model 2	CAR	0.859 (0.810–0.909)	< 0.001*
Model 3	NLR	0.861 (0.812–0.910)	< 0.001*
Model 4	IL–6	0.852 (0.802–0.903)	< 0.001*
Model 5	hs-CRP	0.850 (0.799–0.901)	< 0.001*
Model 6	PNI	0.844 (0.792–0.897)	< 0.001*

Note: *$P < 0.05$; statistically significant. All models adjusted for the same covariates as in Table 5

Table 7 Incremental predictive value of biomarkers for double VC and any CVC

Model	Variable	Outcome	Continuous NRI (95% CI)	*P*-value	IDI (95% CI)	*P*-value
Model 1	CALLY index	Double VC	0.435 (0.174–0.695)	0.001*	0.034 (0.011–0.058)	0.004*
		Any CVC	0.573 (0.298–0.848)	< 0.001*	0.059 (0.019–0.100)	0.004*
Model 2	CAR	Double VC	0.088 (-0.204–0.380)	0.555	0.005 (-0.005–0.015)	0.350
		Any CVC	0.356 (0.070–0.643)	0.015*	0.039 (0.009–0.070)	0.012*
Model 3	NLR	Double VC	0.190 (-0.099–0.479)	0.197	0.009 (-0.005–0.023)	0.223
		Any CVC	0.282 (-0.007–0.571)	0.056	0.011 (-0.012–0.034)	0.350
Model 4	IL-6	Double VC	0.069 (-0.209–0.348)	0.625	0.003 (-0.005–0.011)	0.504
		Any CVC	0.403 (0.126–0.679)	0.004*	0.020 (-0.003–0.042)	0.084
Model 5	hs-CRP	Double VC	-0.075 (-0.367–0.216)	0.614	0.001 (-0.003–0.006)	0.584
		Any CVC	0.134 (-0.156–0.425)	0.365	0.016 (-0.005–0.038)	0.146
Model 6	PNI	Double VC	0.414 (0.127–0.701)	0.005*	0.026 (0.003–0.050)	0.029*
		Any CVC	0.503 (0.218–0.787)	< 0.001*	0.001 (-0.002–0.005)	0.408

Note: *$P < 0.05$; statistically significant. All models adjusted for age, sex, dialysis vintage, serum phosphorus, phosphate binder type, and active vitamin D dosage

(NRI = 0.435, $P = 0.001$) and an integrated discrimination improvement (IDI = 0.034, $P = 0.004$) for differentiating double valve calcification from no calcification. The PNI also showed significant improvements for the same endpoint (NRI = 0.414, $P = 0.005$; IDI = 0.026, $P = 0.029$). In contrast, for double valve calcification, CAR, NLR, IL-6, and hs-CRP failed to yield statistically significant improvements in either continuous NRI or IDI (all $P > 0.05$) (Table 7).

Stratified bootstrap validation (2000 resamples) confirmed the robustness of the main findings. The CALLY index, NLR, and CAR maintained significant associations with double valve calcification, with minimal bias and BCa confidence intervals excluding zero (CALLY: bias – 0.071, BCa 95% CI:-1.087–-0.397, $P < 0.001$; NLR: bias 0.022, 0.047–0.746, $P = 0.010$; CAR: bias 0.100, 0.253–2.030, $P = 0.002$). In contrast, the originally significant association of PNI became non-significant after bootstrap correction (bias – 0.027, $P = 0.095$), while IL-6, hs-CRP, and GNRI did not consistently show significance.

Bayesian multinomial logistic regression yielded results that were materially consistent with both the maximum likelihood and bootstrap estimates. For the CALLY index, the posterior mean coefficient for double valve calcification was – 0.70 (95%CI: -1.11–-0.35), corresponding to a posterior odds ratio of approximately 0.50. The posterior estimates for NLR (0.38, 0.11–0.68) and CAR (1.01, 0.35–1.71) also remained significant. PNI showed only a marginal posterior effect (-0.09, -0.18–-0.01), while IL-6, hs-CRP, and GNRI exhibited credible intervals that crossed or approached zero. All models showed excellent convergence (Rhat = 1.00, bulk-ESS > 2000).

Subgroup analyses

The CALLY index independently predicted both aortic valve calcification (AVC) (OR = 0.594, $P < 0.001$) and mitral valve calcification (MVC) (OR = 0.744, $P = 0.018$). Notably, hs-CRP, NLR, and CAR only predicted AVC (all $P < 0.05$), but not MVC (all $P > 0.05$) (Table 8).

ROC curve analysis

ROC analysis showed that the CALLY index had the largest AUC (0.751, 95%CI: 0.682–0.820) for predicting CVC, outperforming IL-6 (0.715), CAR (0.732), hs-CRP (0.681), and NLR (0.643). The detailed parameters of the ROC analysis are presented in Table 9, while the ROC curves are illustrated in Fig. 4.

Discussion

In this cross-sectional study of 203 MHD patients, we systematically evaluated the association between microinflammatory state and CVC, and for the first time compared the predictive value of the CALLY index with other inflammatory and nutritional biomarkers for CVC

Table 8 Logistic regression analysis of aortic valve calcification and mitral valve calcification

Variable	AVC (*n* = 117)		MVC (*n* = 87)	
	OR (95% CI)	*P*-value	OR (95% CI)	*P*-value
hs-CRP	1.193 (1.023–1.390)	0.024*	1.013 (0.889–1.154)	0.847
IL–6	1.070 (0.978–1.172)	0.139	1.010 (0.975–1.047)	0.577
NLR	1.281 (1.032–1.590)	0.025*	1.096 (0.935–1.285)	0.257
CAR	2.248 (1.350–3.742)	0.002*	1.111 (0.754–1.638)	0.595
CALLY index	0.594 (0.454–0.777)	< 0.001*	0.744 (0.583–0.950)	0.018*
PNI	0.998 (0.976–1.020)	0.846	0.921 (0.856–0.991)	0.029*
GNRI	1.000 (0.984–1.015)	0.963	0.965 (0.920–1.012)	0.140

Note: *$P < 0.05$; statistically significant. CAR was analyzed after being standardized for each increase of 0.1. The models were adjusted for age, gender, dialysis vintage, phosphorus level, type of phosphate binder, active vitamin D dosage, and history of diabetes mellitus and coronary artery disease.

Table 9 ROC curve parameter values of hs-CRP, PNI, GNRI, NLR, CAR, IL-6, and CALLY index for predicting CVC

Variable	AUC (95% CI)	Sensitivity / Specificity	Cut-off Value	*P*-Value
hs-CRP(mg/L)	0.681 (0.601–0.762)	75.0% / 56.7%	2.920	< 0.001*
IL–6(pg/mL)	0.715 (0.632–0.798)	78.7% / 64.2%	4.490	< 0.001*
NLR	0.643 (0.563–0.723)	72.1% / 50.7%	3.119	< 0.001*
CAR	0.732 (0.659–0.806)	94.1% / 41.8%	0.029	< 0.001*
CALLY index	0.751 (0.682–0.820)	80.9% / 56.7%	1.707	< 0.001*
PNI	0.678 (0.604–0.753)	35.3% / 92.5%	40.575	< 0.001*
GNRI	0.648 (0.569–0.727)	44.1% / 80.6%	94.224	< 0.001*

Note: *$P < 0.05$; statistically significant. The cut-off value represents the optimal threshold determined by the Youden index

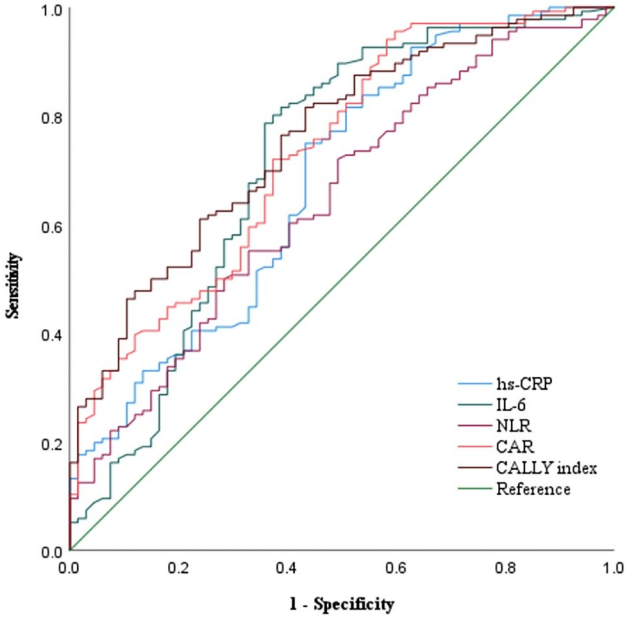


Fig. 4 ROC of hs-CRP, NLR, CAR, IL-6,CALLY index in MHD Patients. The AUC for the CALLY index, CAR, IL-6 was 0.751 ,0.732 and 0.715 ($P < 0.001$), while the AUC for hs-CRP and NLR was 0.681 and 0.643 ($P < 0.001$)

severity. The main findings are as follows. The CALLY index showed a stepwise decrease with increasing CVC severity and remained an independent protective factor after adjusting for multiple confounders. The CALLY showed the best model fit and discrimination among the markers assessed. Non-calcium-containing phosphate binders and active vitamin D were independently associated with reduced calcification odds. The CALLY index independently predicted both aortic and mitral valve calcification, whereas hs-CRP, NLR, and CAR only predicted aortic valve calcification.

The present study demonstrated that phosphorus and calcium-phosphorus product were independent risk factors for CVC, with increasing severity of CVC associated with higher levels. This finding aligns with the established role of calcium-phosphorus homeostasis in cardiovascular calcification. Vieceli Dalla Sega et al. emphasized that hyperphosphatemia-induced osteogenic differentiation of vascular smooth muscle cells represents a core mechanism of cardiovascular calcification, with elevated calcium-phosphorus product directly promoting ectopic calcium deposition in valvular tissue [6]. Juhász et al., in a prospective study of 154 cardiovascular patients, first demonstrated that the plasma phosphate/pyrophosphate ratio was significantly associated with aortic valve calcification burden, remaining an independent determinant even after adjusting for traditional risk factors, highlighting the central role of phosphate homeostasis dysregulation in valvular calcification [39]. Vervloet systematically summarized the impact of CKD-mineral and bone disorder on arterial stiffness, noting that hypercalcemia and hyperphosphatemia promote vascular calcification and increase arterial stiffness, thereby exacerbating cardiovascular risk [40].

Non-calcium-containing phosphate binder use and higher active vitamin D dosage were associated with lower odds of double valve calcification. These findings are consistent with accumulating evidence on the

cardiovascular benefits of optimizing phosphate binder selection. Mason et al., in a systematic evaluation, demonstrated that calcium-containing phosphate binders promote vascular calcification progression more significantly than sevelamer, supporting the preferential use of non-calcium-containing agents [41]. Zhang et al., in a meta-analysis of 3,271 patients, confirmed that sevelamer was associated with reduced coronary artery calcification progression compared with calcium-based binders [42]. Regarding vitamin D analogs, Gravellone et al. summarized that active vitamin D receptor activators exert cardiovascular protective effects through suppression of PTH secretion, upregulation of Klotho expression, and enhancement of matrix Gla protein activity [43]. More recently, Meng et al. demonstrated in animal models that the phosphate binder sevelamer significantly attenuates bioprosthetic valve calcification through mechanisms independent of serum calcium and phosphorus regulation [44].

A key aspect of this work is the direct comparison of the CALLY index with PNI and GNRI, which are established nutritional-inflammatory indices. Yaprak et al. recently demonstrated that GNRI, MIS, and PNI predict mortality in hemodialysis patients, with GNRI performing best [18]. In our data, however, PNI and GNRI did not exhibit robust independent associations with CVC severity, and PNI lost significance after bootstrap validation. This discrepancy may be due to the absence of a direct inflammatory component (CRP) in PNI and GNRI, whereas the CALLY index intentionally incorporates CRP, thereby better capturing the low-grade inflammation that drives valvular calcification [6, 12–15]. Similarly, Wang et al. showed that dynamic changes in PNI and GNRI were associated with reduced mortality [19]; our findings extend this concept to structural cardiovascular endpoints, suggesting that for CVC, inflammatory load is a crucial dimension.

The CALLY index also demonstrated favorable incremental predictive value, significantly improving both continuous NRI and IDI for double valve calcification. Although CAR and NLR showed significant associations in some models, only the CALLY index consistently improved both metrics. These results echo the concept advanced by Arslan that a multidimensional assessment of cardiometabolic risk—integrating nutrition, inflammation, and body composition—may outperform simpler markers [20]. The CALLY index, derived from widely available laboratory parameters, offers a pragmatic summary of several interrelated pathways.

The differential predictive pattern for aortic versus mitral valve calcification merits consideration. The CALLY index predicted both, whereas single-dimension inflammatory markers predicted only aortic calcification. This aligns with reports that mitral valve calcification is more strongly influenced by metabolic and nutritional factors [45, 46] and underscores the value of a composite marker for comprehensive risk assessment.

Although serum TNF-α levels differed across the three CVC groups in unadjusted comparisons, this association was no longer significant after multivariable adjustment. This pattern may reflect confounding by age and other inflammatory markers; in a recent study of 58 maintenance hemodialysis patients, IL-6 demonstrated stronger and more consistent associations with echocardiographic remodeling parameters than TNF-α, which was more strongly influenced by age and albumin levels [47]. An earlier report found that TNF-α was an independent risk factor for CVC; however, that analysis did not simultaneously adjust for IL-6, CRP, and other inflammatory indices [48]. These observations suggest that when multiple inflammatory markers are concurrently measured and adjusted, as in the present study, TNF-α offers limited additional predictive value beyond IL-6 and composite markers such as the CALLY index.

Several limitations should be acknowledged. First, the cross-sectional design precludes causal inference, and the temporal relationship between inflammation and valve calcification remains uncertain. Second, the relatively modest sample size may limit generalizability. These analyses were limited by substantially reduced sample sizes, which may result in low statistical power and imprecise confidence intervals. Third, echocardiographic assessment of CVC was qualitative; although two experienced sonographers independently acquired images, no formal inter-observer agreement statistics were calculated, and computed tomography-based calcium scoring was unavailable for quantification. Fourth, inflammatory markers were measured at a single time point, which may not fully capture the dynamic nature of microinflammation. Additionally, potential residual confounding from dialysis adequacy (Kt/V was comparable across groups), medication adherence (documented from prescription records rather than objectively assessed), and other unmeasured factors cannot be excluded. Future multicenter prospective studies with longitudinal follow-up are warranted to validate these findings.

Conclusion

In summary, microinflammatory state, as reflected by circulating inflammatory biomarkers, was independently associated with the severity of CVC. The CALLY index was independently associated with both aortic and mitral valve calcification and showed favorable predictive performance for assessing CVC severity in this population.

Abbreviations

CALLY	C-reactive protein-albumin-lymphocyte
CVC	Cardiac valvular calcification
MHD	Maintenance hemodialysis

ROC	Receiver operating characteristic
hs-CRP	High-sensitivity C-reactive protein
IL-6	Interleukin-6
CKD	Chronic kidney disease
ALP	Alkaline phosphatase
Alb	Serum albumin
BUN	Blood urea nitrogen
SCr	Serum creatinine
UA	Uric acid
TC	Total cholesterol
TG	Triglycerides
HDL-c	High-density lipoprotein cholesterol
LDL-c	Low-density lipoprotein cholesterol
iPTH	Intact parathyroid hormone
WBC	White blood cell count
PLT	Platelet count
Hb	Hemoglobin
TNF-α	Tumour necrosis factor
CAD	Coronary artery disease
AHA/ACC	American Heart Association/American College of Cardiology
DM	Diabetes mellitus
ADA	American Diabetes Association
CDC	Centers for Disease Control and Prevention
IQR	Interquartile range
AUC	Area under the curve
CI	Confidence intervals
BMI	Body mass index
LAD	Left atrial diameter
LVEF	Left ventricular ejection fraction
NLR	Neutrophil-to-lymphocyte ratio
CAR	C-reactive protein-to-albumin ratio
Cac	Corrected calcium
Ca×P	Calcium-phosphorus product
Single VC	Single valve calcification
Double VC	Double valve calcification group
PNI	Prognostic Nutritional Index
GNRI	Geriatric Nutritional Risk Index
RRF	Residual renal function
BCa	Bias-corrected
NRI	Net reclassification improvement
IDI	Integrated discrimination improvement

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Author contributions
The research idea and study design were conceived by Wei Lu. Peishan Peng contributed to data collection. Meiyu Liu was responsible for study design, data collection, analysis and interpretation, and drafting the initial manuscript. Jie Chen and Wei Lu critically reviewed and revised the manuscript. All authors have read and approved the final version of the manuscript.

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Data availability
The data underlying this article will be shared on reasonable request to the corresponding author.

Declarations

Ethical approval and consent to participate
Ethical approval was obtained from the Ethics Committee of Xinhua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine (Approval No: XHEC-C-2019-087). The study was performed in compliance with the 1964 Declaration of Helsinki. All data were fully anonymised and analysed without individual identifiers. Written informed consent was obtained from all patients.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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